

3(a) Using $v = f\lambda$

$$v = \frac{1}{4.0 \times 10^{-3}} \times 1.4$$

$$= 350 \text{ m s}^{-1}$$

M1

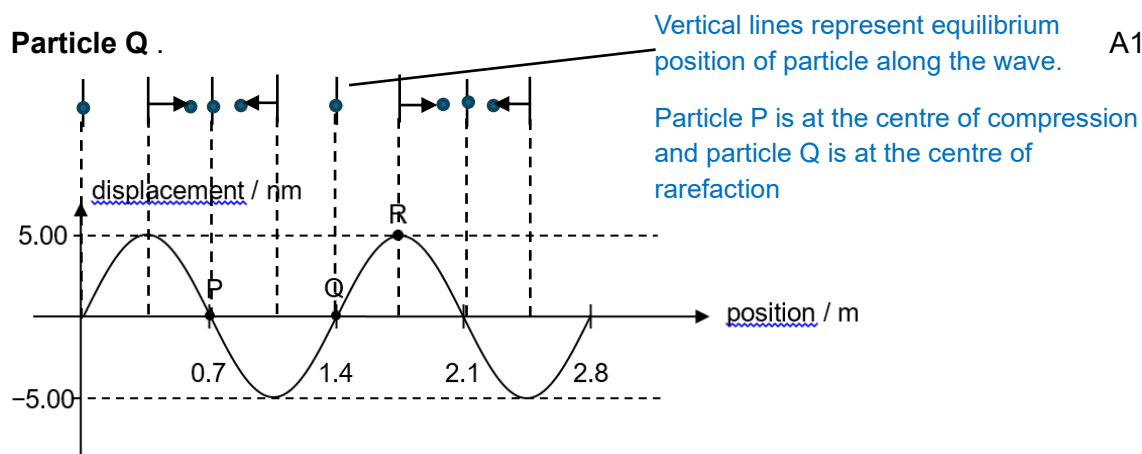
A1

(b)(i) Particle R

(As the particle is undergoing SHM, at the amplitude, the instantaneous velocity of the particle is zero).

A1

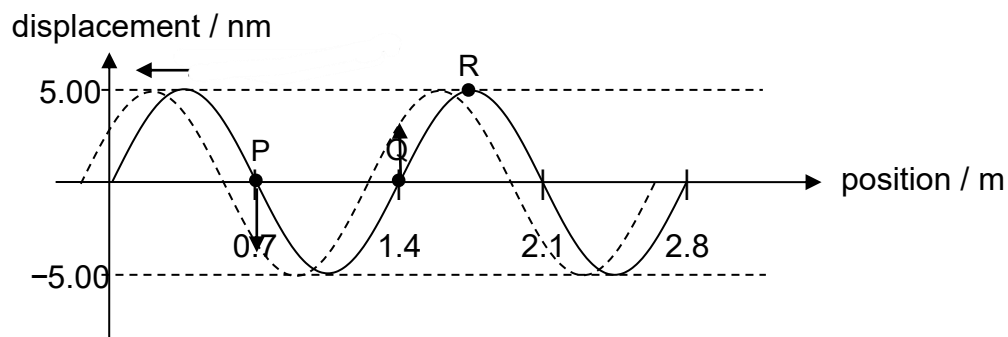
(b)(ii) Particle Q .



A1

(b)(iii) Displacement of article Q will be positive next instant.

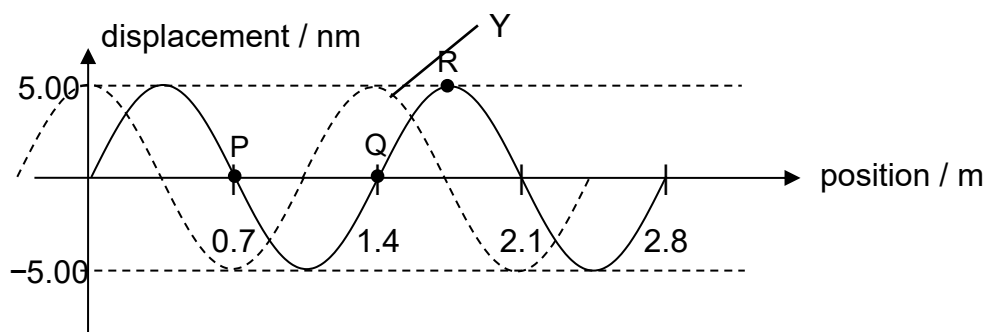
A1



(c)(i) Distance travelled by wave in 1 ms = $vt = (350)(1 \times 10^{-3}) = 0.35$ m

The graph should have shifted to the left by 0.35 m.

C1



A1

Award mark as long as one full wavelength is drawn with displacement at 5.00 nm at initial position.

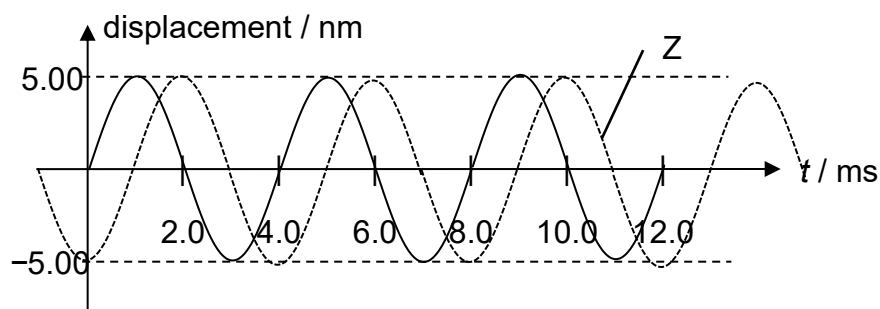
(c)(ii) Phase difference between particle R and S

C1

$$\Delta\phi = \frac{\Delta x}{\lambda} \times 360^\circ = \frac{0.7}{1.4} \times 360^\circ = 180^\circ$$

OR

Phase difference between particle Q and S is 90° .



A1

